

Moukthika Nellutla

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EDUCATION

University of Illinois Urbana-Champaign

Expected December 2027

Bachelor of Computer Science and Bioengineering

GPA: 3.51/4.00

- **Relevant Coursework:** Introduction to Computer Science, Data Structures and Algorithms, Molecular and Cellular Biology, Linear Algebra, Database Systems, Computational Tools for Biological Data, Machine Learning, Bioinformatics, Signals and Systems, Differential Equations
- **Certifications:** Break Through Tech Cornell University online certificate in Machine Learning Foundations *July 2026*

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript, HTML/CSS, R, MATLAB, SQL

Frameworks: React, Flask, FastAPI, Node.js, TypeScript, Agile, Lambda, Operating Systems, Angular, PHP

Developer Tools: Git, Docker, Jupyter, VS Code, GCP, AWS, Linux, OAuth, DynamoDB, CloudFormation, Vue, Azure

Libraries: NumPy, PyTorch, scikit-learn, OpenCV, Pandas, IoT, TensorFlow, SciPy

Applications: PostgreSQL, SQLite, Excel, LabVIEW, SharePoint, Android, Greenfield, API Gateway, Terraform, Spark, Snowflake, Jenkins, DevOps, Maven, Ray, Kubernetes, Tableau, SVN, REST API, Stripe Checkout

EXPERIENCE

Software Engineer

October 2025 – May 2026

Gies College of Business

Champaign, IL

- Architected an LLM inference platform used by **10+** faculty members for academic evaluation, building secure **REST APIs**, scalable backend services, **interactive analytics projection**, and product management dashboards.
- Optimized model inference pipelines for latency and computation efficiency, increasing business system reliability and throughput by **30%**.
- Streamlined **RAG** workflows with automated logging, supporting **100s** of LLM test runs.

Machine Learning Intern

May 2025 – August 2025

Moffitt Cancer Center

Tampa, FL

- Deployed NLP and **cross-functional machine learning techniques** to enhance clinical trial matching for **20k+ oncology patients** using **Deep6 AI** and Jupyter model workflows.
- Benchmarked **6+ ML models** (MLP, Logistic Regression, Gradient Boosting, KNN, SVM) using **confusion matrices** and performance metrics for data automation.
- Engineered scalable preprocessing and feature engineering pipelines that supported **100s** of patient-trial matches for downstream model training.

COMMUNITY & LEADERSHIP

Undergraduate Research Assistant

April 2025 – Present

Sweedler Research Group

Champaign, IL

- Quantifying **5k+ individual peptides** in rat dorsal root ganglia to study neuropathic pain using **LC-MS/MS data**.
- Authored Python/R scripts to **group overlapping** peptides by sequence position, sum signals, and differentiate unique vs. shared peptides.
- Planned **statistical comparisons via data science** across three animal groups (D, H, J) during acute pain stage to **identify biomarkers**.

Project Manager

Sept 2024 – Present

NeuroTech@UIUC

Champaign, IL

- Formulated Python ML models to analyze EEG signals, correlating **15k+ channel peaks** with cognitive tasks and enhancing preprocessing pipelines.
- Spearheaded NeuroBot, merging EEG preprocessing with **transformer-based NLP prompt** pipelines to generate personalized dashboards, **leading a 10-15 member team**.
- Implemented real-time EEG data acquisition modules with artifact rejection and visualization dashboards to support ongoing experiments.

PROJECTS

Demynsify | Python, FastAPI, NLP, Machine Learning

Dec 2025 – Jan 2026

GitHub: github.com/Mnellutla1120/demynsify

- Curated a **full-stack medical misinformation detection system** to classify health claims as reliable or misleading with **100,000+** articles, papers, etc. from 1997-2022.
- Designed preprocessing pipelines for medical text and trained supervised models on curated public-health datasets.

MediPay | Python, FastAPI, React, SQLAlchemy, Stripe API

Feb 2026 – Mar 2026

GitHub: github.com/Mnellutla1120/hackillinois2026medicaldebt

- Devised a **full-stack REST API** (FastAPI, SQLAlchemy, Pydantic) at **HackIllinois 2026** that scores medical debt risk and generates interest-adjusted repayment plans with configurable down payments.
- Automated **Stripe Checkout** for monthly, down-payment, and custom payments, and exposed **8+ CRUD/query** endpoints with auto-generated **OpenAPI/Swagger documentation**.